

Haley Colgate Kottler, PhD(c)

Data Scientist

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Education and Certifications

University of Wisconsin - Madison Madison, WI	
PhD in Mathematics	expected May 2026
Delta Research Mentor badge	May 2025
Erdős Institute Data Science Bootcamp	April 2025
Delta Teaching & Learning Practitioner badge	Dec 2024
MA in Mathematics	2022
Colorado College Colorado Springs, CO	
BA in Mathematics with Distinction, cum laude	2020

Skills Summary

Creative and detail-oriented problem solver, committed to leveraging data analysis methods to improve health care processes and patient outcomes in collaboration with a cross-functional team.

Programming in R, Python, Julia, SQL, LaTeX | Data visualization | Written and verbal communication | Presenting and communicating technical information to diverse audiences | Adaptability | Mentorship and teaching | Linux CLI | Causal inference | Machine learning

Professional Experience

Lead Data Analyst BerbeeWalsh Department of Emergency Medicine Madison, WI Jan	2024–present
- Designed and implemented data processing pipelines for novel multilevel data - Developed study design and data analysis plans with preliminary data for grant applications - Cleaned, analyzed, and presented data as a part of a cross-disciplinary team of clinicians, researchers, and policy experts to prepare publications and guide future research directions - Mentored three junior colleagues to determine appropriate statistical techniques and presentations - Visualized, explained, and presented data analysis results for a wide audience	
Mathematics Instructor University of Wisconsin-Madison Madison, WI	2023
- Designed a syllabus, assessments, lecture notes, and course policies for an accelerated summer semester class on ordinary differential equations - Created lectures and in class activities for 30 students in each of two developmental math classes - Developed scaffolded exercises and corresponding assessments to track progress and assess understanding - Communicated key ideas and techniques with a focus on helping students develop an intuition for why methods work and when they are appropriate to use	
Mathematics Teaching Assistant University of Wisconsin-Madison Madison, WI	2020-2022
- Established a resource bank for course materials to encourage sharing materials and coordinated with colleagues to streamline workflows - Provided constructive feedback to students to promote recognition of areas for growth - Advised junior colleagues on determining and addressing the root cause of why students struggle with a concept and on conflict resolution strategies for classroom management - Engaged students in the curriculum by describing the practical uses of technical concepts to make content relevant for non-mathematics major students	

Leadership Experience

- Organizer and problem writer | Mega Math Meet | Madison, WI 2023-present
Directed 12 graders for a mathematics competition for 180 fifth and sixth grade students each year from around the state of Wisconsin, computing scores and awards rankings during the morning of the competition as students completed team and individual problems we developed based on advanced concepts not included in the usual K12 math curriculum
- Graduate Math Program Committee | University of Wisconsin-Madison 2023–present
Gathered, collated, and presented the wishes of the mathematics graduate student body to department leadership to guide improvement and future directions of the program
- Co-founder | UGMMAW | University of Wisconsin-Madison 2023–2025
Established a community among undergraduate students studying mathematics who may not feel included by hosting social events and workshops, to create an environment where they can mingle, find support from their peers, and advance themselves professionally

Awards and Fellowships

- Exceptional Service Award | University of Wisconsin-Madison, Mathematics Dept. 2024
This award is given to PhD students who have performed exceptional service related to the mission of their department and university.
- Institute for Foundations of Data Science Fellowship | University of Wisconsin-Madison 2023
This fellowship is given to PhD students to work on research topics that fit well with the IFDS goal of promoting transdisciplinary research in the fundamentals of data science.
- Early Excellence TA Award | University of Wisconsin-Madison, Mathematics Dept. 2022
This award is given to TAs who have demonstrated excellence in teaching over three or fewer semesters.

Selected Communications

Publications

- [A simplified and robust proxy-based approach for overcoming unmeasured confounding in EHR studies](#)
arXiv pre-print June 2025
- [Characterizing Lucid Episodes in Advanced Dementia: A Prospective Study](#)
The Gerontologist Aug 2025
- [Quantifying aberrant approach-avoidance conflict in psychopathology: A review of computational approaches](#)
Neuroscience & Biobehavioral Reviews April 2023
- [Batch Codes from Affine Cartesian Codes and Quotient Spaces](#)
Cryptography and Coding - 18th IMA International Conference Dec 2021
- [Colorado in context: Congressional redistricting and competing fairness criteria in Colorado](#)
Journal of Computational Social Science May 2021

Software

- GMMPParameterEstimation.jl is a Julia package that uses method of moments for parameter recovery for Gaussian Mixture Models. <https://github.com/HaleyColgateKottler/GMMPParameterEstimation.jl>